

Exponential Inequalities for Reversible Markov chains with Spectral Gap

Abstract

We derive concentration inequalities for stationary additive functionals of a finite reversible Markov chain. Comparing the transition matrix with a stay-or-refresh chain reduces the moment generating function bound to one equation for the largest eigenvalue of a tilted matrix. For bounded observables, a Hoeffding-type bound follows; with knowledge of the variance, an improved Bennett-type bound becomes available. The variance inflation factor is uniformly sharp, while the Bernstein scale is optimal within the rational envelope used here.

1 Introduction

Concentration inequalities describe how a random quantity fluctuates around its typical value. For sums of independent bounded random variables, two of the basic examples are Hoeffding's and Bernstein's inequalities: Hoeffding's inequality uses only the range of the summands, while Bernstein's inequality can be sharper when their variance is small. Both ultimately come from controlling the moment generating function of the sum in question.

For a Markov chain, the analogous problem is to understand an additive functional

$$S_n = \sum_{k=1}^n f(X_k),$$

where successive terms are no longer independent. One might expect that this requires detailed information about the transition mechanism of the chain. A useful feature of *reversible* chains is that a spectral gap already gives a surprisingly effective summary of their dependence. The main idea of this note is to exploit this through a particularly simple comparison process.

Suppose that P is the transition operator of a stationary reversible chain and that its nonconstant spectrum is bounded above by $\lambda < 1$. Then, in quadratic-form order,

$$P \preceq Q := \lambda I + (1 - \lambda)\Pi,$$

where Π replaces the current state by an independent draw from stationarity. The comparison chain Q therefore has a simple interpretation: at each step it stays where it is with probability λ , and with probability $1 - \lambda$ it forgets its past and moves to an independent draw from π . Its dependence comes in geometric stretches of repeated states in between refreshment events. This elementary model turns out to dominate the original chain for the moment-generating function estimates of interest here.

After exponentially tilting by the observable f , controlling the Markov chain MGF translates into bounding the operator norm of a suitable linear operator. For the stay-or-refresh chain, this operator has enough simplifying structure that its largest eigenvalue can be described exactly by a scalar equation. The concentration problem is thereby reduced to bounding the root of a one-dimensional function. Feeding the usual scalar Hoeffding estimate

into this equation gives a Markov chain Hoeffding inequality. Feeding in a variance-sensitive Bennett estimate instead produces a corresponding Bennett-type bound; choosing a simple analytic envelope then yields a Bernstein inequality.

The same comparison philosophy has a continuous-time analogue. There, the spectral gap allows the generator to be compared with the generator of the corresponding refreshment process, and the tilted spectral problem again collapses to a scalar resolvent estimate. This produces a continuous-time Bernstein inequality with the same limiting constants obtained by shrinking the sampling interval in discrete time.

The emphasis here is on the proof mechanism rather than on maximal generality. The arguments are elementary once the operator reduction has been made, and they provide a useful example of how spectral information about a Markov chain can be converted into explicit non-asymptotic probabilistic estimates. Closely related results are already available in the existing literature, sometimes in greater generality and with somewhat different constants; the aim here is mainly to isolate a simple, self-contained argument and make the underlying comparison mechanism transparent.

2 Setting and main bounds

Let $(X_k)_{k \geq 0}$ be a stationary, irreducible, reversible Markov chain on a finite state space E , with transition matrix P and invariant distribution π . For a real-valued observable f , write

$$\pi(f) := \sum_{x \in E} f(x) \pi(x), \quad S_n := \sum_{k=1}^n f(X_k).$$

For simplicity, we assume throughout that f is centered under π , i.e. $\pi(f) = 0$.

Expectations and probabilities with subscript π refer to the stationary chain. We assume $n \geq 1$ and $|E| \geq 2$; the one-state case is trivial. List the eigenvalues of P as

$$1 = \lambda_1 > \lambda_2 \geq \dots \geq \lambda_{|E|} \geq -1,$$

and set

$$\lambda := \max\{\lambda_2, 0\}, \quad \gamma := 1 - \lambda, \quad A_\lambda := \frac{1 + \lambda}{1 - \lambda}, \quad B_\lambda := \frac{1 + 4\lambda + \lambda^2}{3(1 - \lambda^2)}. \quad (1)$$

Only the upper nonconstant eigenvalue enters the bounds; an absolute spectral gap is not required. Moreover, when $\lambda_2 < 0$, γ is the truncated gap used for the comparison, rather than the exact gap $1 - \lambda_2$.

For $a \leq f \leq b$, we prove

$$\log \mathbb{E}_\pi \left[e^{tS_n} \right] \leq \frac{nA_\lambda (b - a)^2 t^2}{8}, \quad t \in \mathbf{R}. \quad (2)$$

If $f \leq c$ with $c > 0$ and $\sigma^2 := \pi(f^2)$, then we also prove

$$\log \mathbb{E}_\pi \left[e^{tS_n} \right] \leq \frac{nA_\lambda \sigma^2 t^2}{2(1 - cB_\lambda t)}, \quad 0 \leq t < \frac{1}{cB_\lambda}. \quad (3)$$

The first bound uses only the range, while the second retains the variance. Near the origin, their quadratic coefficients are respectively $nA_\lambda(b - a)^2/8$ and $nA_\lambda\sigma^2/2$; as such, when both sets of assumptions are available, the variance-sensitive bound always weakly improves the quadratic coefficient.

For $\lambda = 0$, the constants reduce to the independent Hoeffding and Bernstein constants. This includes the independent chain $P = \Pi$, but does not require independence.

The proof has three steps. First, the spectral gap assumption lets us replace P with a canonical ‘stay-or-refresh’ chain which is particularly tractable. Second, the largest eigenvalue of the tilted comparison matrix satisfies a scalar equation. Third, we bound the root of that equation using the available information about f .

The comparison follows the approach of León and Perron [2]; general-state-space developments appear in Fan, Jiang and Sun [3]. Related operator methods for Bernstein inequalities are developed by Jiang, Sun and Fan [4]. The purpose of this note is to give a self-contained finite-state argument with explicit constants, together with its continuous-time counterpart.

3 Spectral comparison and a scalar eigenvalue equation

Equip $L^2(\pi)$ with the inner product

$$\langle g, h \rangle_\pi := \sum_{x \in E} g(x) h(x) \pi(x), \quad \|g\|_\pi^2 := \langle g, g \rangle_\pi.$$

Let $\mathbf{1}$ denote the constant function equal to one, and define

$$\Pi g := \pi(g) \mathbf{1}, \quad Q := \lambda I + \gamma \Pi.$$

The chain with transition matrix Q stays at its current state with probability λ and draws afresh from π with probability γ . Reversibility and the definition of λ give the quadratic-form inequality

$$P \preceq Q. \tag{4}$$

Indeed, if $g = \pi(g) \mathbf{1} + g_0$ with $\pi(g_0) = 0$, then

$$\langle g, Pg \rangle_\pi = \pi(g)^2 + \langle g_0, Pg_0 \rangle_\pi \leq \pi(g)^2 + \lambda \|g_0\|_\pi^2 = \langle g, Qg \rangle_\pi.$$

For $t \in \mathbf{R}$, define $D_t g(x) := e^{tf(x)/2} g(x)$ and

$$\rho_f(t) := \|D_t Q D_t\|_{L^2(\pi) \rightarrow L^2(\pi)}.$$

Stationarity gives the identity

$$\mathbb{E}_\pi \left[e^{tS_n} \right] = \left\langle D_t \mathbf{1}, (D_t P D_t)^{n-1} D_t \mathbf{1} \right\rangle_\pi. \tag{5}$$

Lemma 1 (Operator reduction). *For every $t \in \mathbf{R}$,*

$$\mathbb{E}_\pi \left[e^{tS_n} \right] \leq \rho_f(t)^n. \tag{6}$$

Proof. Conjugation by the diagonal matrix with entries $\sqrt{\pi(x)}$ turns $D_t P D_t$ into a symmetric matrix with nonnegative entries. For such a matrix, taking the coordinatewise absolute value of a vector cannot decrease the absolute value of its Rayleigh quotient. Perron–Frobenius implies that its operator norm equals its largest eigenvalue, even when P has negative eigenvalues. By (4),

$$D_t P D_t \preceq D_t Q D_t, \quad \|D_t P D_t\| \leq \rho_f(t).$$

Consequently, (5) is at most $\pi(e^{tf}) \rho_f(t)^{n-1}$. The Rayleigh quotient of $D_t Q D_t$ at $D_t \mathbf{1}$ gives

$$\rho_f(t) \geq \frac{\lambda \pi(e^{2tf}) + \gamma \pi(e^{tf})^2}{\pi(e^{tf})} \geq \frac{(\lambda + \gamma) \pi(e^{tf})^2}{\pi(e^{tf})} = \pi(e^{tf}),$$

where the second inequality is Cauchy–Schwarz; we conclude. $\square$

3.1 Bounding the exponential growth rate

Let $M_{e^{tf}}$ denote multiplication by e^{tf} . The positive matrix

$$T_t := QM_{e^{tf}} = D_t^{-1} (D_t Q D_t) D_t$$

has Perron eigenvalue $\rho = \rho_f(t)$ and a positive eigenfunction h . Writing $m := \pi(e^{tf}h)$, its eigenvalue equation becomes

$$\rho h(x) = \lambda e^{tf(x)} h(x) + \gamma m, \quad h(x) = \frac{\gamma m}{\rho - \lambda e^{tf(x)}}.$$

In particular, $\rho > \lambda \max_x e^{tf(x)}$, and substitution into the definition of m yields

$$1 = \gamma \pi \left(\frac{e^{tf}}{\rho - \lambda e^{tf}} \right). \quad (7)$$

For $r > \lambda \max_x e^{tf(x)}$, define

$$G_{f,t}(r) := \gamma \pi \left(\frac{e^{tf}}{r - \lambda e^{tf}} \right).$$

If $\lambda > 0$, this function decreases strictly from infinity to zero as r runs through its admissible interval, so this equation determines ρ uniquely. When $\lambda = 0$, it equals $\pi(e^{tf})/r$, and the uniqueness of the solution persists.

This gives a direct comparison rule: to bound the eigenvalue, it is enough to find an admissible r at which $G_{f,t}(r) \leq 1$. Since $G_{f,t}$ decreases and $G_{f,t}(\rho_f(t)) = 1$, it holds that

$$G_{f,t}(r) \leq 1 \implies \rho_f(t) \leq r. \quad (8)$$

The rest of the discrete-time proof uses this rule to compare scalar quantities.

4 The Hoeffding bound

Theorem 2 (Markov–Hoeffding). *If $a \leq f \leq b$ with $a < b$ and $\pi(f) = 0$, then (2) holds. For every $u > 0$,*

$$\mathbb{P}_\pi(S_n \geq u) \leq \exp \left\{ -\frac{2u^2}{nA_\lambda(b-a)^2} \right\}. \quad (9)$$

The same estimate holds for $\mathbb{P}_\pi(S_n \leq -u)$.

4.1 Reduction to a two-state chain

For fixed $r > \lambda \max\{e^{ta}, e^{tb}\}$, the function

$$\phi_r(y) := \frac{e^{ty}}{r - \lambda e^{ty}}$$

is convex on $[a, b]$, since

$$\phi_r''(y) = \frac{t^2 r e^{ty} (r + \lambda e^{ty})}{(r - \lambda e^{ty})^3} \geq 0.$$

For $\mu := (\pi(f) - a) / (b - a)$, the chord inequality for a convex function gives

$$\pi(\phi_r(f)) \leq (1 - \mu) \phi_r(a) + \mu \phi_r(b). \quad (10)$$

Assume first that $0 < \mu < 1$. Let Y take the two values a and b with probabilities $1 - \mu$ and μ , respectively, so that

$$\mathbb{E}Y = \pi(f), \quad (1 - \mu)\phi_r(a) + \mu\phi_r(b) = \mathbb{E}\phi_r(Y).$$

Let ρ_Y be the Perron root for this two-state observable, solving $\gamma\mathbb{E}\phi_{\rho_Y}(Y) = 1$. Its admissibility for the endpoint values also makes it admissible for $f \in [a, b]$. At this root, the endpoint inequality and the scalar comparison rule give

$$\begin{aligned} \pi(\phi_{\rho_Y}(f)) &\leq (1 - \mu)\phi_{\rho_Y}(a) + \mu\phi_{\rho_Y}(b) = \mathbb{E}\phi_{\rho_Y}(Y) = \gamma^{-1} \\ \implies G_{f,t}(\rho_Y) &\leq 1 \\ \implies \rho_f(t) &\leq \rho_Y. \end{aligned}$$

As a result, it suffices to prove the desired bound for the two-state chain. Finally, if $\mu \in \{0, 1\}$, then $a \leq f \leq b$ and $\pi(f) = (1 - \mu)a + \mu b$ force f to be constant almost-surely; centering then makes $S_n = 0$ almost-surely, so there is nothing to prove.

Lemma 3 (Two-state estimate). *Let $a < b$, $0 < \mu < 1$, and $0 \leq \lambda < 1$. Set*

$$Q_\lambda = \lambda I + (1 - \lambda) \begin{pmatrix} 1 - \mu & \mu \\ 1 - \mu & \mu \end{pmatrix}, \quad D_t = \text{diag}(e^{ta/2}, e^{tb/2}).$$

Let $\theta(t)$ be the largest eigenvalue of $D_t Q_\lambda D_t$. With

$$m = (1 - \mu)a + \mu b, \quad d = b - a, \quad A_\lambda = \frac{1 + \lambda}{1 - \lambda},$$

we have, for every $t \in \mathbb{R}$,

$$\log \theta(t) \leq mt + \frac{A_\lambda d^2 t^2}{8}.$$

Proof. We first record the elementary two-point form of Hoeffding's lemma. For $s \in [0, 1]$, let $L_s(t) = \log((1 - s)e^{ta} + se^{tb})$. Writing $s_t = se^{tb} / ((1 - s)e^{ta} + se^{tb})$, differentiation gives $L_s''(t) = d^2 s_t(1 - s_t) \leq d^2/4$. Since $L_s(0) = 0$ and $L_s'(0) = (1 - s)a + sb$, integrating this second-derivative bound yields

$$(1 - s)e^{ta} + se^{tb} \leq \exp\left(t[(1 - s)a + sb] + \frac{d^2 t^2}{8}\right) \quad (t \in \mathbb{R}).$$

If $\lambda = 0$, the rows of Q_0 both equal $(1 - \mu, \mu)$. Its tilted matrix has largest eigenvalue $\theta(t) = (1 - \mu)e^{ta} + \mu e^{tb}$, so the preceding two-point inequality, with $s = \mu$, proves the claim. Suppose now that $0 < \lambda < 1$.

The matrices $D_t Q_\lambda D_t$ and $Q_\lambda D_t^2$ are similar, with determinant $\lambda e^{t(a+b)}$. Define $p = \frac{\lambda + (1 - \lambda)\mu}{1 + \lambda}$. A direct calculation computes their trace as $(1 + \lambda)[(1 - p)e^{ta} + pe^{tb}]$. Writing then $M_p(t) = (1 - p)e^{ta} + pe^{tb}$, their eigenvalues are the (positive, real) roots of

$$q(z) := z^2 - (1 + \lambda)M_p(t)z + \lambda e^{t(a+b)} = 0.$$

Put $\omega = A_\lambda$, $z_* = \exp\left(mt + \frac{\omega d^2 t^2}{8}\right)$. We claim first that $q(z_*) \geq 0$. Because $z_* > 0$, this is equivalent to

$$\frac{z_* + \lambda e^{t(a+b)} z_*^{-1}}{1 + \lambda} \geq M_p(t).$$

Set $U = \log z_*$ and $V = t(a + b) - U$. Convexity of the exponential gives

$$\frac{e^U + \lambda e^V}{1 + \lambda} \geq \exp\left(\frac{U + \lambda V}{1 + \lambda}\right).$$

The definitions above imply the equalities

$$\frac{m + \lambda(a + b - m)}{1 + \lambda} = (1 - p)a + pb, \quad \frac{1 - \lambda}{1 + \lambda} \omega = 1,$$

and the last exponential is hence equal to

$$\exp\left(t[(1 - p)a + pb] + \frac{d^2 t^2}{8}\right),$$

which is at least $M_p(t)$ by the two-point inequality above. This proves the claimed inequality and hence $q(z_*) \geq 0$.

It remains to check that z_* lies *above* the larger root of the characteristic polynomial, rather than below the smaller one. Completing the square gives

$$\log \frac{z_*^2}{e^{t(a+b)}} = (2\mu - 1)dt + \frac{\omega d^2 t^2}{4} \geq -\frac{(2\mu - 1)^2}{\omega} \geq -\frac{1}{\omega}.$$

Also,

$$\log \lambda \leq \lambda - 1 \leq -\frac{1 - \lambda}{1 + \lambda} = -\frac{1}{\omega}.$$

Consequently $z_*^2 \geq \lambda e^{t(a+b)}$, the product of the two roots, and z_* is at least their geometric mean. Since $q(z_*) \geq 0$, it follows that z_* is at or above the larger root, namely $\theta(t)$. Taking logarithms, we conclude. $\square$

Proof of Theorem 2. Combine the endpoint reduction, Lemma 3 with $m = 0$, and Lemma 1. This gives (2). Chernoff's inequality with $t = 4u / [nA_\lambda(b - a)^2]$ gives (9); apply the same argument to $-f$ for the lower tail. $\square$

5 A variance-sensitive Bennett bound

Assume throughout this section that

$$\pi(f) = 0, \quad f \leq c, \quad c > 0, \quad \pi(f^2) = \sigma^2. \quad (11)$$

Let $X \sim \pi$ and put $Z := f(X)/c$ and $v := \sigma^2/c^2$, so that $\mathbb{E}[Z] = 0$, $Z \leq 1$, and $\mathbb{E}[Z^2] = v$. Define $\psi(s) := e^s - 1 - s$. For $s \geq 0$,

$$\mathbb{E}[e^{sZ}] \leq 1 + v\psi(s). \quad (12)$$

Indeed, Taylor's integral formula gives, for $z \leq 1$,

$$e^{sz} - 1 - sz = s^2 z^2 \int_0^1 (1 - u) e^{usz} du \leq z^2 (e^s - 1 - s),$$

and averaging removes the linear term.

The denominator in the scalar eigenvalue equation can be expanded as a geometric series, with terms naturally indexed by the number of consecutive steps before a refreshment occurs. To this end, let K have the geometric distribution

$$\mathbb{P}(K = k) = (1 - \lambda) \lambda^{k-1}, \quad k \geq 1,$$

with $K = 1$ almost surely when $\lambda = 0$. Then $\mathbb{E}[K] = 1/(1 - \lambda)$. Define the ‘dependence-adjusted Bennett function’ H_λ by

$$H_\lambda(a) := (1 - \lambda) \mathbb{E}[\psi(aK)] = \frac{(1 - \lambda)^2 e^a}{1 - \lambda e^a} - (1 - \lambda) - a, \quad 0 \leq a < -\log \lambda. \quad (13)$$

Here and below $-\log 0 := +\infty$, so that $H_0(a) = e^a - 1 - a$ for all $a \geq 0$.

Theorem 4 (Bennett bound). *Under (11),*

$$\log \mathbb{E}_\pi \left[e^{tS_n} \right] \leq n \frac{\sigma^2}{c^2} H_\lambda(ct), \quad 0 \leq ct < -\log \lambda. \quad (14)$$

Proof. Write $a := ct$, $g := vH_\lambda(a)$, and $r := e^g$. Since $\lambda e^a < 1 \leq r$, r is admissible. By (8), it is enough to show that

$$G_{f,t}(e^g) = (1 - \lambda) \mathbb{E} \left[\frac{e^{aZ}}{e^g - \lambda e^{aZ}} \right] \leq 1.$$

Expanding the denominator and applying (12) gives

$$\begin{aligned} G_{f,t}(e^g) &= (1 - \lambda) \sum_{k \geq 1} \lambda^{k-1} e^{-gk} \mathbb{E} \left[e^{akZ} \right] \\ &\leq \mathbb{E} \left[e^{-gK} \right] + v \mathbb{E} \left[e^{-gK} \psi(aK) \right]. \end{aligned} \quad (15)$$

All terms are nonnegative, so the expansion and interchange of expectations are justified by monotone convergence. Since $1 - e^{-x} \geq xe^{-x}$ for $x \geq 0$,

$$1 - \mathbb{E} \left[e^{-gK} \right] \geq g \mathbb{E} \left[K e^{-gK} \right]. \quad (16)$$

In view of (15), it therefore remains to control the second term by

$$v \mathbb{E} \left[e^{-gK} \psi(aK) \right] \leq g \mathbb{E} \left[K e^{-gK} \right].$$

Since $g = vH_\lambda(a)$, this is equivalent to

$$\frac{\mathbb{E} \left[e^{-gK} \psi(aK) \right]}{\mathbb{E} \left[K e^{-gK} \right]} \leq H_\lambda(a).$$

The factor K in the denominator naturally suggests passing to the size-biased law. Accordingly, introduce the size-biased law $\nu(k) := kP(K = k)/\mathbb{E}[K]$. The function $k \mapsto \psi(ak)/k$ is increasing: for real $x > 0$, the derivative of $\psi(ax)/x$ has the sign of $(ax - 1)e^{ax} + 1 \geq 0$; by contrast, $k \mapsto e^{-gk}$ is decreasing. Their covariance under ν is therefore nonpositive, so it follows that

$$\mathbb{E}_\nu \left[e^{-gK} \frac{\psi(aK)}{K} \right] \leq \mathbb{E}_\nu \left[e^{-gK} \right] \mathbb{E}_\nu \left[\frac{\psi(aK)}{K} \right].$$

By the definition of the size-biased law,

$$\mathbb{E}_\nu \left[e^{-gK} \frac{\psi(aK)}{K} \right] = \frac{\mathbb{E} \left[e^{-gK} \psi(aK) \right]}{\mathbb{E}[K]},$$

while

$$\mathbb{E}_\nu \left[e^{-gK} \right] = \frac{\mathbb{E} \left[K e^{-gK} \right]}{\mathbb{E}[K]}, \quad \mathbb{E}_\nu \left[\frac{\psi(aK)}{K} \right] = \frac{\mathbb{E}[\psi(aK)]}{\mathbb{E}[K]}.$$

Substituting these identities and cancelling gives

$$\frac{\mathbb{E} \left[e^{-gK} \psi(aK) \right]}{\mathbb{E} [K e^{-gK}]} \leq \frac{\mathbb{E} [\psi(aK)]}{\mathbb{E} [K]} = H_\lambda(a). \quad (17)$$

Multiplying by $v\mathbb{E} [K e^{-gK}]$ bounds the second term in (15) by $g\mathbb{E} [K e^{-gK}]$. Equation (16) now gives $G_{f,t}(e^g) \leq 1$. Thus $\rho_f(t) \leq e^g$, and Lemma 1 proves the result. $\square$

The geometric variable K corresponds to the length of a block before refreshment occurs, with mean $1/(1-\lambda)$ equal to the persistence time of the comparison chain. Formula (13) expresses the effect of dependence through the Bennett function evaluated at this random block length.

6 The Bernstein envelope and tail bound

The coefficients in (1) satisfy

$$H_\lambda(0) = H'_\lambda(0) = 0, \quad H''_\lambda(0) = A_\lambda, \quad H'''_\lambda(0) = 3A_\lambda B_\lambda.$$

Consequently,

$$H_\lambda(a) = \frac{A_\lambda a^2}{2} + \frac{A_\lambda B_\lambda a^3}{2} + O(a^4) \quad \text{as } a \downarrow 0. \quad (18)$$

Any rational upper bound of the form

$$H_\lambda(a) \leq \frac{A_\lambda a^2}{2(1 - Ba)}$$

which is valid near zero must therefore have $B \geq B_\lambda$. We now show that the limiting value $B = B_\lambda$ does give an upper bound.

Lemma 5 (Optimal sub-gamma envelope). *For $0 \leq a < B_\lambda^{-1}$,*

$$H_\lambda(a) \leq \frac{A_\lambda a^2}{2(1 - B_\lambda a)}. \quad (19)$$

Among envelopes of this form with quadratic coefficient A_λ , the scale B_λ is minimal.

Proof. Set $R_\lambda(a) := A_\lambda a^2 / [2(1 - B_\lambda a)]$. The two functions agree in value and first derivative at zero. It is therefore enough to compare their second derivatives:

$$H''_\lambda(a) = \frac{(1-\lambda)^2 e^a (1 + \lambda e^a)}{(1 - \lambda e^a)^3}, \quad R''_\lambda(a) = \frac{A_\lambda}{(1 - B_\lambda a)^3}. \quad (20)$$

The cubic denominator in R''_λ suggests taking an inverse cube root: on any interval where $1 - B_\lambda a > 0$, the desired inequality $H''_\lambda(a) \leq R''_\lambda(a)$ is equivalent to the statement that

$$\left(\frac{A_\lambda}{H''_\lambda(a)} \right)^{1/3} \geq 1 - B_\lambda a.$$

Define then

$$F_\lambda(a) := \left(\frac{A_\lambda}{H''_\lambda(a)} \right)^{1/3} = \frac{1 - \lambda e^a}{1 - \lambda} \left[\frac{1 + \lambda}{e^a (1 + \lambda e^a)} \right]^{1/3}. \quad (21)$$

For $q := \lambda e^a$ and $s(q) := (1 + 4q + q^2) / [3(1 - q^2)]$, logarithmic differentiation gives

$$-\frac{F'_\lambda(a)}{F_\lambda(a)} = s(q), \quad \frac{F''_\lambda(a)}{F_\lambda(a)} = s(q)^2 - qs'(q) = \frac{(1-q)^2}{9(1+q)^2} \geq 0,$$

exhibiting that F_λ is convex, with $F_\lambda(0) = 1$ and $F'_\lambda(0) = -B_\lambda$.

Considering its tangent at $a = 0$ therefore yields $F_\lambda(a) \geq 1 - B_\lambda a$ for $0 \leq a < -\log \lambda$. For $0 < \lambda < 1$, we have $F_\lambda(a) \rightarrow 0$ as $a \uparrow -\log \lambda$. Hence

$$0 \geq 1 - B_\lambda(-\log \lambda) \implies B_\lambda^{-1} \leq -\log \lambda.$$

For $\lambda = 0$, the domain restriction is automatic. Consequently, for $0 \leq a < B_\lambda^{-1}$, we have $1 - B_\lambda a > 0$, and the tangent inequality is equivalent to

$$H''_\lambda(a) \leq R''_\lambda(a).$$

Integrating twice proves (19); minimality follows from (18). $\square$

Remark 6 (Independent constants). When $\lambda = 0$, $A_0 = 1$, $B_0 = 1/3$, and $F_0(a) = e^{-a/3}$. The curvature argument reduces to $e^{-a/3} \geq 1 - a/3$, giving the usual Bernstein envelope for $e^a - 1 - a$.

Theorem 7 (Markov–Bernstein). *Under (11), let $V_\lambda := A_\lambda \sigma^2$ and $b_\lambda := cB_\lambda$. Then (3) holds, and for every $u > 0$,*

$$\mathbb{P}_\pi(S_n \geq u) \leq \exp \left\{ -\frac{u^2}{2(nV_\lambda + b_\lambda u)} \right\}. \quad (22)$$

If also $|f| \leq c$, then

$$\mathbb{P}_\pi(|S_n| \geq u) \leq 2 \exp \left\{ -\frac{u^2}{2(nV_\lambda + b_\lambda u)} \right\}. \quad (23)$$

Proof. Theorem 4 and Lemma 5 give

$$\log \mathbb{E}_\pi [e^{tS_n}] \leq n \frac{\sigma^2}{c^2} H_\lambda(ct) \leq \frac{nV_\lambda t^2}{2(1 - b_\lambda t)}.$$

If $\sigma^2 = 0$, $f = 0$ and both tail bounds are immediate. Otherwise take $t = u / (nV_\lambda + b_\lambda u) < b_\lambda^{-1}$ in Chernoff's inequality. This gives (22). Applying it to $-f$ and adding the two tails gives (23). $\square$

The moment bound is sub-gamma on the right, with variance parameter nV_λ and scale b_λ . A two-sided sub-gamma statement requires the corresponding bound for $-f$, as ensured by $|f| \leq c$.

Remark 8. The preceding discrete-time arguments are not restricted to finite state spaces. Let P be the Markov operator of a stationary reversible chain on a general measurable state space, acting self-adjointly on $L^2(\pi)$, and suppose that

$$\langle h, Ph \rangle_\pi \leq \lambda \|h\|_2^2, \quad h \perp \mathbf{1}.$$

Equivalently,

$$P \preceq \lambda I + (1 - \lambda)\Pi$$

in quadratic-form order on $L^2(\pi)$.

For bounded measurable f , the same comparison extends to general state spaces. The finite-state derivation above uses Perron–Frobenius to obtain the scalar root equation; in the general setting this step is replaced by the corresponding rank-one resolvent or variational characterization. The subsequent scalar estimates are unchanged, and the same MGF and concentration bounds follow.

7 Interpretation of the constants

For the stationary stay-or-refresh chain and a centred observable with variance σ^2 ,

$$\text{Cov}_\pi(f(X_0), f(X_k)) = \lambda^k \sigma^2.$$

Thus, for every $n \geq 1$,

$$\text{Var}_\pi(S_n) = \sigma^2 \left[n + 2 \sum_{k=1}^{n-1} (n-k) \lambda^k \right] = n A_\lambda \sigma^2 - \frac{2\sigma^2 \lambda (1 - \lambda^n)}{(1 - \lambda)^2}. \quad (24)$$

In particular,

$$n^{-1} \text{Var}_\pi(S_n) \rightarrow A_\lambda \sigma^2.$$

Indeed, any uniform MGF bound with quadratic coefficient C ,

$$\log \mathbb{E}_\pi[e^{tS_n}] \leq \frac{nC\sigma^2 t^2}{2} + o(t^2) \quad \text{as } t \downarrow 0,$$

necessarily implies $\text{Var}_\pi(S_n) \leq nC\sigma^2$, and so no smaller coefficient than A_λ can replace the variance inflation factor uniformly over all n and all chains with this spectral parameter. The symmetric two-state observable attains $\sigma^2 = (b-a)^2/4$, which also establishes the uniform sharpness of the coefficient in the Hoeffding bound.

The optimality statement for B_λ has a different scope: it is the smallest scale of a rational envelope of H_λ that retains the quadratic coefficient A_λ . This follows from the cubic expansion and Lemma 5. Of course, the Bennett estimate itself may contain some slack, and so this loosening may also.

The asymptotic behaviour in the limit of small spectral gap is explicit. With $\gamma = 1 - \lambda$,

$$A_\lambda = \frac{2}{\gamma} - 1, \quad B_\lambda = \frac{1}{\gamma} - \frac{1}{2} - \frac{\gamma}{6(2-\gamma)}. \quad (25)$$

Both the variance and scale are inflated by order γ^{-1} , in agreement with the typical length of a block in between refreshments.

8 Continuous time

Let $(X_s)_{s \geq 0}$ now be a stationary reversible Markov process with invariant distribution π and a strongly continuous self-adjoint Markov semigroup on $L^2(\pi)$. We take a jointly measurable version of the process, so that $(s, \omega) \mapsto X_s(\omega)$ is measurable. Throughout, integrals of the form $\int_0^T h(X_s) ds$ denote ordinary Lebesgue integrals in the time variable; in particular, they are well defined for bounded measurable h .

Write L for the generator of the process and assume the spectral gap inequality

$$-\langle g, Lg \rangle_\pi \geq \gamma \text{Var}_\pi(g), \quad g \in \mathcal{D}(L), \quad \gamma > 0. \quad (26)$$

Let f be bounded and measurable, with $\pi(f) = 0$, $f \leq c$, $c > 0$, and $\pi(f^2) = \sigma^2$. Set

$$A_T := \int_0^T f(X_s) ds, \quad T > 0.$$

Theorem 9 (Continuous-time Bernstein bound). *Under these assumptions,*

$$\log \mathbb{E}_\pi[e^{tA_T}] \leq \frac{T\sigma^2 t^2}{\gamma - ct}, \quad 0 \leq t < \frac{\gamma}{c}. \quad (27)$$

Consequently, for every $u > 0$,

$$P_\pi(A_T \geq u) \leq \exp \left\{ -\frac{\gamma u^2}{4T\sigma^2 + 2cu} \right\}. \quad (28)$$

If $|f| \leq c$, the two-sided bound holds with a factor of two.

Proof. Let M_f denote multiplication by f . The bounded-potential Feynman–Kac formula and the spectral theorem give

$$E_\pi \left[e^{tA_T} \right] = \left\langle \mathbf{1}, e^{T(L+tM_f)} \mathbf{1} \right\rangle_\pi \leq \exp \{ T \sup \text{spec} (L + tM_f) \}.$$

To make the comparison precise, let $\mathcal{E}(h, h) := \|(-L)^{1/2}h\|_2^2$, $\mathcal{D}(\mathcal{E}) := \mathcal{D}((-L)^{1/2})$. The spectral-gap inequality (26) extends by closure from $\mathcal{D}(L)$ to $\mathcal{D}(\mathcal{E})$, giving

$$\mathcal{E}(h, h) \geq \gamma(\|h\|_2^2 - \pi(h)^2), \quad h \in \mathcal{D}(\mathcal{E}).$$

Thus, with $L_0 := -\gamma(I - \Pi)$, the associated quadratic forms satisfy

$$-\mathcal{E}(h, h) \leq \langle h, L_0 h \rangle_\pi, \quad h \in \mathcal{D}(\mathcal{E}).$$

Since f is bounded, the perturbation by tM_f preserves the form domain, and hence

$$-\mathcal{E}(h, h) + t\pi(fh^2) \leq \langle h, (L_0 + tM_f)h \rangle_\pi, \quad h \in \mathcal{D}(\mathcal{E}).$$

Taking Rayleigh quotients therefore yields

$$\sup \text{spec}(L + tM_f) \leq \sup \text{spec}(L_0 + tM_f),$$

so it suffices to bound the latter spectral supremum.

Fix $r \geq 0$ with $r + \gamma > tc$ and let $w := r + \gamma - tf > 0$. Weighted Cauchy–Schwarz gives, for every real $h \in L^2(\pi)$,

$$\pi(h)^2 \leq \pi(w^{-1}) \pi(wh^2).$$

It follows that

$$\gamma\pi\left(\frac{1}{r + \gamma - tf}\right) \leq 1 \implies \langle h, [rI - (L_0 + tM_f)]h \rangle_\pi = \pi(wh^2) - \gamma\pi(h)^2 \geq 0. \quad (29)$$

Obtaining a scalar bound on $\pi(w^{-1})$ thus gives an upper bound on the spectrum through the quadratic form; note that this argument does not require the spectral supremum to be an exact eigenvalue.

To verify (29), write $d := r + \gamma$. The identity

$$\frac{1}{d - tf} = \frac{1}{d} + \frac{tf}{d^2} + \frac{t^2 f^2}{d^2(d - tf)}$$

and $f \leq c$ yield

$$\pi\left(\frac{1}{d - tf}\right) \leq \frac{1}{d} + \frac{t^2 \sigma^2}{d^2(d - tc)}. \quad (30)$$

Choose $r := \sigma^2 t^2 / (\gamma - ct)$ for $0 \leq t < \gamma/c$. Since $d \geq \gamma$ and $d - tc \geq \gamma - tc$, it follows that

$$\frac{\gamma t^2 \sigma^2}{d(d - tc)} \leq \frac{\sigma^2 t^2}{\gamma - tc} = r,$$

and hence (30) implies $\gamma\pi(w^{-1}) \leq (\gamma + r)/d = 1$. Equation (29) gives $L_0 + tM_f \preceq rI$, proving (27).

The right-hand side of (27) is a sub-gamma bound with variance parameter $2T\sigma^2/\gamma$ and scale c/γ . The Chernoff calculation in Theorem 7 gives (28), with the zero-variance case handled separately as before. $\square$

Related spectral and Bernstein concentration estimates for continuous-time Markov processes appear in Lezaud [5] and Gao, Guillin and Wu [6].

8.1 Discrete-to-continuous scaling

For a finite-state continuous-time chain, sampling at spacing $\Delta > 0$ gives a transition matrix whose nonconstant spectrum is bounded above by $\lambda_\Delta := e^{-\gamma\Delta}$. The discrete proof uses only such an upper bound, so it applies with λ_Δ . For $T = n\Delta$, the Riemann sum $\Delta \sum_{k=1}^n f(X_{k\Delta})$ has sub-gamma parameters

$$T\Delta A_{\lambda_\Delta}\sigma^2, \quad c\Delta B_{\lambda_\Delta}.$$

As $\Delta \downarrow 0$ with T/Δ integral,

$$\Delta A_{\lambda_\Delta} \longrightarrow \frac{2}{\gamma}, \quad \Delta B_{\lambda_\Delta} \longrightarrow \frac{1}{\gamma}.$$

These converge to the parameters in Theorem 9. In the finite-state setting, sample paths have finitely many jumps on bounded intervals almost surely, so the Riemann sums converge to A_T ; boundedness of f also allows passage to their moment generating functions. The direct proof above avoids needing such a sampling argument on a general state space.

The discrete and continuous arguments share the same structure: compare the original dynamics with a refreshed process, then use elementary estimates on the observable to bound the tilted operator. In discrete time this reduces to comparing the root of a decreasing scalar function; in continuous time weighted Cauchy–Schwarz gives the corresponding quadratic form bound.

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